White paper: the pmsimstats-ng manuscript compendium

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A five-page review of the eleven manuscripts under `analysis/report/`, with an assessment of the contribution each makes and a prioritized set of next steps.

Scope and method of this review

The compendium comprises eleven manuscripts, each in its own numbered directory, all targeting *Statistics in Medicine* style and sharing a common preamble, bibliography apparatus, and simulation substrate. Several directories carry `report-slim`, `report-extra-slim`, `report-mini`, or `report_short` variants; these are length-reduced renderings of the same study, not separate investigations, and are assessed here as one paper each.

This review is source-level. Every `report.Rmd` was read for its abstract, section structure, and results, together with the project READMEs, the six committed referee reports, the paper-04 remediation plan, and the three existing per-paper white papers (01, 02, 03). No simulation was re-executed and no numerical result was independently recomputed for this document, so all quantitative statements below are reported as the manuscripts report them. Assessments of contribution and of weakness are the reviewer's judgment; claims about draft state (placeholder counts, missing results, deferred sweeps) are verified by inspection of the sources.

Update, 2026-08-12. This revision reflects work committed to the compendium between 2026-07-29 and 2026-08-12, established by `git diff` against commit `fdfa40e` (the commit that produced the original text below) and by reading the current working-tree sources. The comparison is source-level in the same sense as the original: no simulation was re-executed for this revision either, so a report that a number is unchanged means the manuscript still states that number, not that it was independently re-derived. Every paragraph below that changed is marked inline; paragraphs with no marker were checked against the current source and found unchanged. Papers 04, 05, and 08 received only copy-edits (British-to-US spelling, an added units footnote reconciling this series' weekly $t_{1/2}$ convention with 04 and 05's day-denominated one) and are otherwise as originally assessed.

What the programme is

The compendium is a single coherent research programme rather than a collection of related papers. Its object is one estimand: the biomarker-by-treatment interaction in aggregated N-of-1 trials, the quantity such trials exist to estimate when they are used to validate a predictive biomarker. Its motivating application throughout is the prazosin/PTSD nightmare-frequency setting of Hendrickson et al. (2020), and its method throughout is Monte Carlo simulation reported under the ADEMP framework of Morris, White, and Crowther (2019).

The eleven papers partition into five layers.

- **Generative substrate.** 06 (three-component BR/PB/TV decomposition) and 07 (Gompertz trajectory template) justify the shape of the simulated data.
- **Interaction encoding.** 01 (Architecture A vs B), 11 (Architecture C, combined), and 03 (latent-class and mixture formulations) ask how the biomarker's moderating effect should enter the data-generating process at all.
- **Analyst-side inference.** 02 (carryover specification A1/A2/A3), 08 (RM-ANOVA vs mixed model vs GEE), and 10 (working-covariance miscalibration and cluster-robust repair) ask what the analyst should fit.
- **Design.** 05 (twelve-sweep design sensitivity), 09 (informative dropout by design family), and the design axis of 08.
- **Clinical framing.** 04 (N-of-1 versus parallel-group RCT for the treatment main effect).

The programme's intellectual center is paper 01. Its finding, that the carryover penalty is an order of magnitude larger when the interaction is encoded in the covariance (Architecture B) than in the mean (Architecture A), propagates as a conditioning variable into 02, 03, 07, and 11, and it is the reason those papers report architecture-stratified rather than pooled conclusions. That is a genuine and unusual structural virtue: the programme has identified a modeling choice that the surrounding literature treats as an implementation detail and has made it an explicit axis everywhere downstream.

Paper-by-paper assessment

01. Two data-generating architectures. *Useful; the keystone paper, and materially strengthened since 2026-07-29.* [Updated.] Establishes that Architecture B loses 30.6% relative power at a one-week carryover half-life in the OL+BDC design against 2.8% for Architecture A, with a difference-of-differences $z = 7.64$, while the crossover design shows no architecture effect; these headline numbers are unchanged. What changed is the paper around them. It was retitled ("Mean moderation and covariance moderation as data-generating architectures…"), rebuilt on `bookdown::pdf_document2` so cross-references resolve, and rewritten end to end: the three-part `bullets` / `rgt` / `orig` scaffold is gone entirely, replaced by continuous author prose with zero remaining placeholder blocks, the first of the eleven papers to reach that state (with 02 and 03, below). The N convention was also corrected to a uniform $N = 70$ total throughout, resolving the per-path-versus-total ambiguity the prior draft carried. Most consequentially, the Architecture C material was fully extracted rather than folded back as this review had recommended (see "Recommended consolidation," updated below); paper 01 now contains no Architecture C content at all, so the duplication this review flagged no longer exists, but the super-additivity benchmark-scale problem travels with the material to paper 11 and is still unresolved there. Section 6's power-recovery strategies remain qualitative hypotheses with no simulation behind them. The directory's `README.md` is now stale in the other direction: it still asserts the `rgt` blocks carry `rgt to complete.` placeholders, which is no longer true of `report.Rmd`.

02. Carryover-mitigation analysis strategy. *Useful; the strongest negative result in the compendium, and the scope has narrowed to its sharpest claim.* [Updated.] The five drafts this review counted (`report`, `-slim`, `-extra-slim`, `-extra-slim-rgt`, `-devresults`) were consolidated on 2026-07-30 into a single master `report.Rmd`, with the others moved to `archive/`; the scaffold is fully retired here too, with zero placeholder blocks remaining. The paper's scope narrowed with it: rather than the full 540-cell, three-specification, three-architecture factorial, the master now restricts to Architecture B, exponential and Weibull decay, and a head-to-head between the exposure-weighted specification (E2, formerly A2) and the lagged-treatment specification (E3, formerly A3) against an unadjusted baseline (E1, formerly A1), filtering to 216 of the original 540 cells. The "19% of non-null cells" framing this review quoted is gone along with the broader grid; in its place the paper states directly that the exposure-weighted advantage is markedly inferior under the classical crossover design (0.488 against 0.830, unchanged from the number this review cited) and should not be extended there. Two of the three weaknesses flagged here are resolved: the CR2 cluster-robust recalibration is now validated across all five S6 stress cells rather than one, holding or widening the exposure-weighted advantage at four of them and narrowing only under high autocorrelation and heavy dropout; and the previously noted inconsistency with paper 01's carryover-loss figures was reconciled before this review was written and remains reconciled, since 01's headline numbers are unchanged. The bias/MSE/coverage cross-specification comparability caveat persists as a structural feature of the design (E1/E3 target a different coefficient than E2), not a defect to be fixed.

03. Latent-class and mixture formulations. *Partially useful; a theory paper with a pilot attached.* Aims 1 and 2 (formalizing the latent-class DGP, deriving when a single MVN substitutes for it, characterizing identifiability) are delivered and are the paper's real contribution, particularly the result that the achievable marginal correlation is bounded by the geometric mean of the two class separations. Aim 3 is a 240-replicate pilot of the last of five pre-registered studies; aim 4, the prazosin re-analysis, has not been attempted. The pilot's findings are sharp (the unconditional class-aware Wald test rejects at 0.197 against nominal 0.05; the BIC-conditional test controls size but is essentially inert at 0.025 power), but they rest on one sample size, one design, and one generative mechanism, and the regime where a class-aware test should win, large separation with genuinely bimodal response, was never tested. [Updated.] The scaffold is now fully written out here as well, zero placeholder blocks remain, and the paper took the reframe-rather-than-rerun path this review offered as an alternative: the abstract and Results now state plainly, in the author's own prose, that the finding holds "in a 240-replicate-per-cell pilot" and "within the 240-replicate pilot," rather than presenting the pilot as if it carried the weight of the pre-registered production run. The canonical 1,000-replicate run at $N \in \{35, 70, 100\}$ over a wider gating-slope grid, including a genuinely bimodal separation regime, has not been executed, so the substantive limitation this review raised is unchanged; what changed is that the manuscript now discloses it correctly rather than needing to.

04. N-of-1 versus parallel-group RCT. *Useful clinically, but carries an unaddressed comparison-fairness problem.* Reports that the aggregated N-of-1 hybrid dominates the parallel RCT across the entire effect-size range, with roughly a threefold power ratio at the moderate-effect cell, negligible bias, and near-nominal coverage in both arms. The ADEMP apparatus is the most complete in the compendium and the 2026-05-19 remediation plan's critical items (dual implementations, RNG discipline) appear to have been carried out; the DerSimonian-Laird estimator now survives only as literature discussion. The problem is the comparison itself. Both arms are matched at $N = 35$ participants, but the N-of-1 arm contributes eight periods per participant against the RCT's single post-baseline measurement, so the result substantially restates that more measurements per subject buy more information. The Limitations section covers compliance, single outcome, and decay form, but does not mention observation count or participant burden at all. A referee at a general medical venue will raise this immediately. Either add a measurement-matched or burden-matched sensitivity arm, or state the asymmetry explicitly and defend it as the decision-relevant comparison for a fixed recruitable population.

05. Twelve-sweep design sensitivity. *Useful as a reference appendix; weak as a standalone paper.* The twelve sweeps (S1-S12) are well chosen, each anchored to a literature precedent, and the null calibration sweep at 2,000 replicates is good practice. The synthesis is defensible: the Hendrickson hybrid is a competent default; ordering flips to the parallel RCT only when carryover exceeds the within-subject period length. But the paper is organized as twelve one-factor-at-a-time sweeps with a single factorial (S11), so it characterizes the axes rather than their interactions, and its target estimand is the main effect, which makes it a companion to 04 rather than to the interaction papers that constitute the programme's core. Its most likely fate is as supplementary material to 04.

06. Three-component decomposition. *Useful; the most intellectually substantial paper after 01.* The omitted-variable-bias identity is the contribution: the one-component biomarker slope is displaced from the pharmacological slope by a term proportional to the biomarker's covariance with the placebo-belief and natural-history components, not by the magnitude of those components. Study A confirms that an uncontaminated biomarker suffers essentially no bias and that the cheap phase-augmented remedy is strictly dominated; the contaminated-biomarker pilot exhibits the failure mode at +0.51 bias; Study B shows recovery is possible only under a balanced-placebo design that decouples exposure from belief. That last result, that the cure is a design property rather than an analysis property, is the paper's real payload and is what the fourth referee report was asking for. Weaknesses: the paper is 26,000 words of source, by a wide margin the longest in the compendium, and needs the slim variant promoted to canonical; the contaminated-biomarker result rests on a 100-replicate pilot while the uncontaminated case gets 1,000; and the practical recommendation (balanced-placebo designs) is rarely feasible in the clinical settings the programme targets, which the Discussion should concede more squarely.

07. Gompertz trajectory evaluation. *Useful, and materially improved by revision.* The first-round referee identified a fatal construction flaw, that the four trajectory families were not four data-generating processes but a post-hoc imposition on Gompertz- generated data. The current draft reports an architecture-dependent answer instead: under Architecture B the trajectory family is exactly inert because the interaction is mean-independent, so matched seeds return identical rejection decisions; under a faithfully multiplicative Architecture A the four families separate by 0.039 in power with Gompertz the mildly conservative member. This is a better paper than the one reviewed, and the exact-inertness result is a clean structural finding worth stating on its own. It remains a narrow paper: one design (OL+BDC), one sample size, zero carryover, 16 cells. The regimes the README promises to identify (non-saturating growth, biphasic patterns, breakpoint behavior) are discussed but not simulated, and the zero-carryover choice is awkward in a programme whose central finding is about carryover. [Checked, unchanged.] The 0.039 power-separation figure, the "exactly inert" result, and the paper's scope (one design, one sample size, zero carryover, 16 cells) are unchanged in the current source; the `rgt` layer still carries 51 placeholder blocks.

08. Test procedure and design. *Useful; the cleanest practical message in the compendium.* The closed-form RM-ANOVA derivation is correct per the referee's independent re-derivation, and the revised simulation now separates the two effects that the first draft confounded: dichotomizing the biomarker costs 0.08 to 0.21 in power, while the further step from a continuous-biomarker contrast to the full mixed model adds at most 0.04. Naive GEE inflates Type I error to 1.5-1.9 times nominal at all four sample sizes and the Mancl-DeRouen correction repairs it at roughly ten points of power. The dichotomization result is the finding most likely to change applied practice, and it deserves to be the paper's headline rather than a component of it. The unresolved problem is scope: the cycle-by-period design grid is specified in detail and then deferred, which leaves the title's second half unearned. Either run the grid or retitle the paper around the test-procedure axis.

09. Informative dropout by design. *Promising, and closer to a paper than it was.* [Updated.] The gap is real and correctly identified: Hendrickson's Figure 4A implies a design-by-dropout coupling that the N-of-1 methods literature has not pursued, and no other manuscript here centers dropout. The abstract's Results paragraph no longer reads "[Forthcoming]" (one internal methods note still uses that word, but the reader-facing Results and Discussion do not), and a `cover-letter.md` has been added. The ceiling-saturation problem this review raised is partly addressed: the design grid was re-run at a matched $N = 70$ and now reports all four designs rather than two, hybrid 0.968, OL+BDC 0.950, traditional crossover 0.854, and open-label 0.088, so the ordering has real resolution between crossover and the top two designs even though hybrid and OL+BDC still sit close to the ceiling and are explicitly flagged in the text as statistically inseparable ($z = 1.44$). The manuscript itself now argues that the hybrid design's earlier near-immunity to dropout was a ceiling artifact of running at half the matched sample size, which is the same diagnosis this review made independently. [Verified 2026-08-12.] Two follow-up items are now checked directly. First, the verbatim-duplicate paragraph this review originally flagged is gone: Results (report.Rmd:947-961) and Discussion's "Prototype Monte Carlo confirmation" subsection (report.Rmd:1304-1328) cover the same prototype run but are no longer the same text; the Discussion version is longer and adds the per-path N breakdown and the contrast with the earlier, unmatched 35-per-path run, so this is no longer a defect. Second, Architecture B and a wider c_{bm} range were not added: the prototype remains single-architecture (mean-moderation only, at one value, $c_{bm} = 0.45$) and covariance-moderation coverage for the OL row is discussed only as a scoped future extension (report.Rmd:1423, 1439, 1561-1575), not run. The `rgt` layer still carries 50 placeholder blocks (still Lorem ipsum throughout the `.rgt` divs, e.g. report.Rmd:944, 1301).

10. Interaction-test calibration. *Useful, and the most portable paper in the compendium.* It is the only manuscript whose contribution is not specific to N-of-1 trials: the finding is that a misspecified parametric working covariance biases the model-based standard error of a mixed-model interaction asymptotically, in either direction. The diagnostic staging is convincing, ruling out degrees-of-freedom rules, positive-definiteness correction, and point-estimate bias, then localizing the 6-10% standard-error inflation to the corCAR1 residual layer, demonstrating that the bias does not attenuate from $N = 35$ to $N = 280$, and repairing calibration with a CR2 cluster-robust standard error and independently from the marginal side with bias-corrected GEE. The reusable staged protocol is a real deliverable. Two gaps: the worked example is the same prazosin cell as everything else, so the generality claim rests on argument rather than on a second application; and the paper has no README and no cover letter, indicating it has not been through the same submission-preparation pass as its neighbors. [Checked, unchanged.] Only cosmetic edits since 2026-07-29: US spelling corrections and relabeling the analysis specifications from A1/A2/A3 to E1/E2/E3 to stay synchronized with paper 02's new naming. Both gaps persist, the `rgt` layer still carries 37 placeholder blocks, and the directory still has no `README.md` or cover letter.

11. Combined DGP architecture (Architecture C). *Still not assessable, and the extraction this review questioned is now complete rather than reversed.* [Updated.] The author did not adopt this review's recommendation to fold 11 back into 01; instead the split was finished. Paper 01's `report.Rmd` no longer contains any Architecture C material, so the duplication this review originally flagged is gone, but that also means paper 11 now carries the entire burden of the material this review found weakest, with none of the consolidation risk addressed. The manuscript's own abstract still states "[To be finalized after the common-driver boundary re-run]", and the super-additivity claim still rests on the language "will confirm agreement to within Monte Carlo standard error" rather than a completed comparison, so the benchmark-scale problem (probability scale versus the effect-size scale this review argued for) has not been resolved. Twelve `rgt` placeholder blocks remain. The crossover-versus-OL+BDC panel question and the non-identifiability of $c_{bm,A}$ and $c_{bm,B}$ from a fitted `bm:Dbc` coefficient were not independently re-checked in this pass.

What the compendium establishes collectively

Read together, the eleven papers deliver a coherent and defensible set of claims about aggregated N-of-1 biomarker-validation trials.

1. How the interaction is encoded in the data-generating process is a scientific assumption with large consequences under carryover, not an implementation detail (01, with 03 and 11 mapping the neighborhood).
2. Analyst-side defaults are conditional, not universal. The exposure-weighted carryover predictor helps only under one architecture and one class of design (02); the mixed model beats GEE and RM-ANOVA, but most of that gap is dichotomization rather than the test (08); the model-based standard error is not trustworthy when the working covariance is a convenient parametric guess (10).
3. Design dominates analysis. Retention beats specification choice (02); the balanced-placebo contrast, not any estimator, is what recovers an unbiased pharmacological slope under a contaminated biomarker (06); design family governs dropout robustness (09).
4. Two modeling choices that might have been feared are largely inert: the trajectory family (07, exactly inert under Architecture
B) and mixture versus continuous encoding at trial-relevant N (03, where the standard mixed model dominates a class-aware test).

Negative and inertness results of this kind are the compendium's comparative advantage, and it is worth being deliberate about that in the cover letters. The programme has repeatedly tested its own defaults and reported when they failed.

Cross-cutting weaknesses

Author prose was absent everywhere; it no longer is, in three papers. [Updated.] At the original baseline all eleven manuscripts used the three-part `bullets` / `rgt` / `orig` scaffold with no author text in any `rgt` block, 645 placeholder blocks in total. As of 2026-08-12, papers 01, 02, and 03 have been rewritten as continuous author prose with the scaffold removed entirely, zero placeholder blocks in each. The remaining eight still carry placeholders: 04 (110), 06 (101), 08 (59), 05 (54), 07 (51), 09 (50), 10 (37), and 11 (12), 474 blocks in total. This is progress on the compendium's single binding submission constraint, but eight of the eleven papers, including paper 10, which this review's next-steps list prioritized third, are unchanged on this dimension.

One reference parameter set. Nearly every simulation is calibrated to the same prazosin/PTSD trajectory. Absolute power values are therefore illustrative throughout, and only orderings port. Paper 10 is the case where this matters most, because its claim is explicitly general.

Ceiling saturation. Papers 09 and, in places, 04 and 05 report power at or near 1.00, where designs and methods cannot be discriminated. Cells should be chosen to sit near 0.5 for the contrast of interest.

Prototype-versus-production gap. Papers 03, 06 (contaminated pilot), 08 (deferred design grid), and 09 present pilots or partial grids as if they carried the weight of the pre-registered production runs. Each states this, but the abstracts do not always reflect it.

Duplicate and stale scaffolding. [Updated.] The READMEs that asserted driver directories "do not yet exist" have been corrected; none remain as of this pass. Paper 09 gained a `cover-letter.md`. Papers 10 and 11 still lack READMEs and cover letters. Paper 01's README is now stale in a new way (see paper 01, above): it still describes the `rgt` layer as unfilled placeholders after the paper was rewritten past that state. Paper 09's Results/Discussion duplication is confirmed resolved: the two sections cover the same prototype run in different prose, not a verbatim repeat.

Recommended consolidation

Eleven manuscripts on a single estimand, calibrated throughout to one prazosin/PTSD parameter set, invites the charge of over-partitioning, and a referee who encounters two of them will notice. Six papers carry a defensible independent claim: 01 (the architecture distinction), 02 (the conditional failure of the exposure-weighted default), 06 (the omitted-variable identity and the design-side cure), 08 (the dichotomization cost), 10 (working-covariance miscalibration, the only contribution not specific to N-of-1 trials), and 04 (the clinical comparison, conditional on repairing its comparison basis). Three do not, and the objection in each case is to the packaging rather than the content, all of which is worth keeping.

- **Fold 11 back into 01** as a scoped sensitivity section rather than completing the extraction. Architecture C is the common generalization of A and B, not a rival to them, and its two channel parameters are not separately identifiable from a fitted `bm:Dbc` coefficient, so it is a simulation-side construct with no estimation counterpart. Its two legitimate uses, a boundary-cell correctness gate on the DGP implementation and a mixing-weight sweep when the mechanism is unknown, are arguments about how to use 01's framework and read naturally as part of 01. **Status as of 2026-08-12: not adopted.** The author completed the extraction instead, and paper 01 is now clean of Architecture C material while paper 11 stands alone, still unfinished, still carrying the unresolved benchmark-scale problem. This recommendation is renewed rather than withdrawn: paper 11 is 3,800-odd words against a compendium median near 13,000, has no README or cover letter, and its abstract still reads as "to be finalized." If the author's judgment is that Architecture C earns independent-manuscript status on scientific grounds, that case should be made explicitly in 11's Introduction, since nothing in the current source argues for it.
- **Publish 05 as supplementary material to 04.** Its twelve sweeps target the treatment main effect, not the interaction, which makes it a companion to 04 rather than to the programme's core, and its one-factor-at-a-time structure characterizes axes rather than their interactions.
- **Merge 07 into 06 as a robustness section on the generative substrate.** Both concern how the simulated trajectories are shaped; neither is large on its own; and 07's cleanest result, that trajectory family is exactly inert under a covariance-channel DGP, is a statement about the substrate that 06 formalizes. If 06 is slimmed for submission as recommended above, 07's sixteen cells fit comfortably in the technical companion.

If adopted, the result would be seven or eight submittable papers of more even weight: 01 (absorbing 11), 02, 04 (with 05 appended), 06 (absorbing 07), 08, 10, and 03 and 09 once their outstanding work completes. As of 2026-08-12 only the 03/09 half of that plan has moved forward on its own terms (03 by reframing, 09 by re-running toward resolvable cells); the 01/11 fold was not adopted, and 05-into-04 and 07-into-06 have not been acted on either.

Recommended next steps, in priority order

1. **Write the `rgt` prose for paper 01, then 02, then 10.** *Status: 01 and 02 done (2026-08-12); 03 done as well, ahead of its place in this list, since it was rewritten in the same pass. Papers 06, 04, 08, 05, 07, 09, and 10 remain, in descending order of remaining placeholder count (101, 110, 59, 54, 51, 50, 37). 10 is still next in priority even though it is not the largest remaining count, because it is the compendium's only non-N-of-1-specific contribution and this review's next-most-portable paper after 01.* Run `~/bin/strip-claudecode` on each as its `rgt` layer closes; 01 and 02 have not yet had it run, since neither has a `README.md` confirming submission-preparation completion in the sense the other finished papers do.
2. **Settle the Architecture C question and re-verify 01's boundary cells.** *Status: not settled, and settled in the opposite direction from this review's recommendation.* The extraction was completed rather than reversed (see "Recommended consolidation," above). What remains outstanding is unchanged: recompute the super-additivity comparison on the effect-size scale, replace the crossover carryover panel with OL+BDC, and confirm that $(c_{bm,A} = 0.45, c_{bm,B} = 0)$ and $(0, 0.45)$ reproduce 01's published power figures exactly before accepting the grid. The 01/02 carryover-loss consistency has already been re-checked and holds, since 01's published numbers did not move.
3. **Fix paper 04's comparison basis.** Add either a measurement-matched or a burden-matched RCT arm, or a stated, defended asymmetry, and put observation count in the Limitations. This is a one-sweep change and it removes the paper's most likely rejection reason.
4. **Re-scope paper 09 around resolvable cells.** *Status: partially done, now verified.* The design-by-dropout grid was re-run at a matched $N = 70$ and now includes all four designs rather than two, giving traditional crossover (0.854) and open-label (0.088) genuine separation from the hybrid/OL+BDC pair, and the "[Forthcoming]" abstract placeholder is gone. The duplicated-paragraph defect this review originally flagged is confirmed removed: Results and Discussion cover the same prototype run in different text, not a verbatim repeat. Still open, and now confirmed rather than merely flagged: Architecture B has not been added (the prototype remains single-architecture, mean-moderation only) and c_{bm} is still evaluated at one value, 0.45, with no range.
5. **Decide paper 08's scope.** Unchanged; no commits to this paper's `report.Rmd` since 2026-07-29 beyond a spelling pass.
6. **Promote slim variants to canonical for 06 and 02.** *Status: done for 02, open for 06.* Paper 02's consolidation folded its slim variant into the single master `report.Rmd` along with the scope narrowing described above, which should bring it within journal budget; this was not independently checked by page count in this pass. Paper 06 is unchanged, still roughly 26,000 words of source against `report-slim.Rmd`'s roughly 18,700, and the slim variant has not been promoted.
7. **Add a second application to paper 10.** Unchanged; no commits to this paper's content since 2026-07-29 beyond a spelling and specification-label pass.
8. **Run the canonical replication for paper 03, or reframe it.** *Status: reframed, not re-run.* The abstract and Results now state directly, in the author's own prose, that the finding holds "in a 240-replicate-per-cell pilot," rather than presenting it as the pre-registered production result. The 1,000-replicate run at $N \in \{35, 70, 100\}$ over a wider gating-slope grid, including a genuinely bimodal separation regime, has not been executed and would still be required before the non-competitiveness result generalizes beyond the pilot.
9. **Housekeeping.** *Status: partially done.* The stale "do not yet exist" README language has been refreshed compendium- wide, and paper 09 gained a cover letter. Still open: READMEs and cover letters for 10 and 11 (11 also still needs the compendium table entry

in `analysis/report/README.md` extended to cover it, not independently re-checked this pass), and paper 01's own README now needs refreshing in the opposite direction, since it undersells a paper that no longer carries `rgt` placeholders.

A reasonable submission sequence, given that the consolidation above was not adopted for paper 11, is 01 first as the keystone (it no longer depends on 11 to be complete), then 02 and 10, both close to complete and 02 already carrying a clean practical message; then 08 once its scope decision is made; then 06 in slim form with 07 as a robustness section, if that consolidation is adopted; then 04 with 05 as supplementary material, once the comparison basis is repaired; with 03, 09, and 11 last, as their outstanding work, reframing accepted for 03, resolution and duplication cleanup for 09, and the benchmark-scale and completion problems for 11, is finished or explicitly closed.